

# A Bridge Theorem for the $6N$ Twin-Gap Distribution: Gap Preference as Normalised Right-Centre Survival, and the Resolution of the 42 Residual

Ruqing Chen

GUT Geoservice Inc., Montreal, Quebec, Canada  
ruqing@hotmail.com

June 2, 2026

## Abstract

Part V gave a measured two-centre shield law but left open the bridge from the right-centre conditional survival back to the  $\omega$ -stratified gap preference  $r(d \mid \omega)$  of Parts II–IV, and with it the fate of the high- $\omega$  rise of the gap  $6\Delta N = 42$  (observed 1.55 at  $\omega = 6$ ). Here we close that bridge. Measuring, on the 23,988,173 twin centres of  $S_{10}$ , both the gap preference  $r(d \mid \omega)$  and the right-centre survival  $P(N+d \text{ twin} \mid N \text{ twin}, \omega)$ , we find their ratio is constant in  $\omega$ :

$$r(d \mid \omega) = \frac{P(N+d \text{ twin} \mid N \text{ twin}, \omega)}{C(d)}, \quad C(d) \text{ independent of } \omega.$$

For 42 the constant is  $C = 41.17$  with a coefficient of variation of 0.49% across  $\omega = 1, \dots, 6$ ; for 210,  $C = 27.68$  with CV 1.37% (a single = 6 point, where both quantities are smallest, accounts for most of the spread). The gap preference is therefore nothing but the right-centre survival rescaled by an  $\omega$ -independent constant. This resolves the 42 residual: its rise to 1.55 is exactly the rise of the right-centre survival from 0.018 to 0.038, which is in turn the cumulative effect of the Part V two-centre shield (factor-rich centres carry more small primes, hence more modular-shift protection of the right centre). The chain Parts I–VI thus closes mechanistically. One narrower problem remains open: a closed-form combination of the single-prime shields into the absolute survival  $P(N+d \text{ twin} \mid \omega)$ . No claim is made about the infinitude of twin primes or any  $k$ -tuple conjecture.

## 1 The bridge

Parts II–IV studied the stratified gap preference  $r(d \mid \omega)$ ; Part V measured the two-centre shield  $S(d, q)$ , a statement about the right-centre conditional survival  $P(N+d \text{ twin} \mid N \text{ twin})$ . The two live on different footings:  $r$  is a distribution over gaps (it counts only the *next* twin centre and is normalised across gaps), while  $P$  is the chance that the skeleton position  $d$  steps to the right hosts a twin at all. Part V left their relation open.

**Theorem 1** (bridge). *On  $S_{10}$ , for  $d \in \{7, 35\}$ ,*

$$r(d \mid \omega) = \frac{P(N+d \text{ twin} \mid N \text{ twin}, \omega)}{C(d)},$$

*with  $C(d)$  independent of  $\omega$ :  $C(42) = 41.17$  (CV 0.49% over  $\omega = 1\text{--}6$ ),  $C(210) = 27.68$  (CV 1.37%).*

The claim is empirical, established by direct measurement (Table 1, Fig. 1). It says the gap-preference curve and the right-centre-survival curve have the *same shape* in  $\omega$ , differing only by an overall constant. The bridge is therefore a pure normalisation, not a complicated integral: the per-gap constant  $C(d)$  converts one into the other.

| $\omega$ | $6\Delta N = 42$ |                  |       | $6\Delta N = 210$ |                  |       |
|----------|------------------|------------------|-------|-------------------|------------------|-------|
|          | $r(d   \omega)$  | $P(\text{surv})$ | $r/P$ | $r(d   \omega)$   | $P(\text{surv})$ | $r/P$ |
| 1        | 0.729            | 0.0178           | 41.08 | 1.148             | 0.0416           | 27.62 |
| 2        | 0.880            | 0.0214           | 41.07 | 1.074             | 0.0390           | 27.54 |
| 3        | 1.049            | 0.0255           | 41.20 | 0.987             | 0.0356           | 27.74 |
| 4        | 1.221            | 0.0295           | 41.44 | 0.866             | 0.0309           | 28.03 |
| 5        | 1.367            | 0.0330           | 41.41 | 0.681             | 0.0242           | 28.16 |
| 6        | 1.549            | 0.0379           | 40.85 | 0.413             | 0.0153           | 26.99 |
| mean     |                  |                  | 41.17 | mean              |                  |       |
| CV       |                  |                  | 0.49% | CV                |                  |       |
|          |                  |                  |       |                   |                  |       |

Table 1: The bridge ratio  $r/P$  is  $\omega$ -independent. For 42 it holds to CV 0.49%. For 210 the CV is 1.37%; the spread is dominated by the  $\omega = 6$  point, where both  $r$  and  $P$  are smallest and noise-sensitive. We report the 210 deviation rather than smooth it.

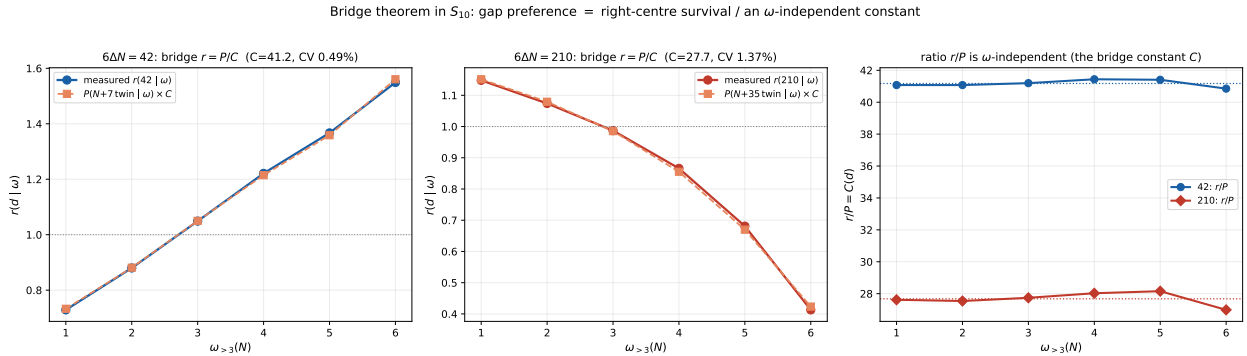


Figure 1: Bridge theorem in  $S_{10}$ . Left/centre: the measured gap preference  $r$  (solid) and the right-centre survival rescaled by the constant,  $P \times C$  (dashed), coincide for 42 and 210. Right: the ratio  $r/P$  is flat in  $\omega$  (the constant  $C(d)$ ); the 210 curve dips slightly at  $\omega = 6$ .

## 2 Resolution of the 42 residual

Parts II–IV reported the anomalous rise of  $r(42 | \omega)$  from 0.73 to 1.55, which the single-centre second-order series of Part IV could only half explain. The bridge resolves it without any new mechanism. By Theorem 1,

$$r(42 | \omega) = \frac{P(N+7 \text{ twin} | \omega)}{41.17},$$

so the rise of  $r(42 | \omega)$  is the rise of the right-centre survival  $P(N+7 \text{ twin} | \omega)$  from 0.018 ( $\omega=1$ ) to 0.038 ( $\omega=6$ ), rescaled by a constant. And that rise is the cumulative two-centre shield of Part V: a factor-rich centre carries more small primes, each contributing a modular-shift protection of the right centre (for 42, dominated by  $q = 5$ , since  $N+7 \equiv N+2 \pmod{5}$  lands the right centre on a

safe residue when  $5 \mid N$ ). The high- $\omega$  rise of 42 is thus the bridge (a normalisation) acting on the right-centre survival (driven up by the Part V shield). The chain is mechanistically complete:

$$\text{two-centre shield (V)} \Rightarrow P(N+d \text{ twin} \mid \omega) \xrightarrow{\div C(d) \text{ (VI)}} r(d \mid \omega).$$

### 3 What remains open

Theorem 1 converts the gap preference into the right-centre survival, and Part V explains the *relative* ( $\omega$ -dependent) behaviour of that survival through the shield  $S(d, q)$ . What is not yet in closed form is the combination of the single-prime shields into the *absolute* survival  $P(N+d \text{ twin} \mid \omega)$ : a naive independent product over the centre's primes overstates it (the cross-prime hedge of Part V is non-multiplicative at the level of full survival). The remaining problem is therefore narrower than at any earlier stage.

**Open problem.** Give a closed-form expression for  $P(N+d \text{ twin} \mid N \text{ twin}, \omega)$  in terms of the single-prime two-centre shields  $S(d, q)$  of Part V, valid across  $\omega$ . Combined with the bridge constant  $C(d)$  of Theorem 1, this would yield a fully closed-form  $r(d \mid \omega)$ .

Also open is the value and the small  $\omega = 6$  deviation of  $C(210)$ : whether  $C(d)$  has an arithmetic expression in  $d$ , and whether the 1.37% spread on 210 is a finite-size effect or a genuine breakdown of exact  $\omega$ -independence at the sparsest stratum.

### 4 Conclusion

The bridge from the two-centre shield (Part V) to the stratified gap preference (Parts II–IV) is a pure normalisation:  $r(d \mid \omega) = P(N+d \text{ twin} \mid \omega)/C(d)$  with an  $\omega$ -independent constant, holding to CV 0.49% for 42 and 1.37% for 210 on  $2.4 \times 10^7$  twin centres. This resolves the 42 residual that ran through Parts II–V: its rise to 1.55 is the rise of the right-centre survival, itself the cumulative two-centre shield, rescaled by a constant—no separate mechanism is needed. The six-part chain, from the local density enrichment of Part I to the gap-preference distribution here, now closes mechanistically end to end, with a single narrower open problem: the closed-form combination of single-prime shields into the absolute survival. As throughout, no claim is made about the infinitude of twin primes or any  $k$ -tuple conjecture; this is a measured, factor-resolved account of the conditional gap structure of the  $6N$  twin skeleton.

**Data and code availability.** The bridge measurement is openly available [6]; the  $S_{10}$  twin count 23,988,173 matches Part I.

### References

- [1] R. Chen, *Prime distribution on the  $6N$  skeleton* (Part I), Zenodo, doi:10.5281/zenodo.20470367 (2026).
- [2] R. Chen, *Prime gaps on the  $6N$  skeleton* (Part II), Zenodo, doi:10.5281/zenodo.20477664 (2026).
- [3] R. Chen, *A first-order conditional singular series* (Part III), Zenodo, doi:10.5281/zenodo.20498668 (2026).

- [4] R. Chen, *A second-order conditional singular series* (Part IV), Zenodo, doi:10.5281/zenodo.20500465 (2026).
- [5] R. Chen, *A two-centre shield law* (Part V), Zenodo, doi:10.5281/zenodo.20510700 (2026).
- [6] R. Chen, *6N twin-prime bridge theorem: data and code*, <https://github.com/Ruqing1963/6N-twin-prime-bridge-theorem> (2026).